REVISED CONSOLIDATED TECHNICAL RESEARCH & NOVELTY REPORT

A Unified Architecture, Empirical Validation & Structural Study of HPP-Net

1. Executive Summary & Core Novelty Formulation

Problem Statement: Low-Light Image Enhancement (LLIE) is bounded by a sharp performance-complexity trade-off. Complex deep networks (e.g., CUE with ~59 Conv2d layers) achieve high fidelity but suffer from prohibitive computation and inference latency on edge platforms. Conversely, ultra-lightweight architectures (e.g., SCI, Zero-DCE) sacrifice parameter representation capacity, resulting in severe noise amplification and color distortions in extremely dark environments due to their lack of spatial context modeling.

HPP-Net Solution: The proposed Hybrid Physical-Prior Network (HPP-Net) resolves this bottleneck. Constrained strictly to **8,153 parameters**, HPP-Net integrates classical Retinex physical model assumptions with a modular Dual Attention structure (composed of parallel Squeeze-and-Excitation channel mapping and multi-scale spatial masking). Optimizing HPP-Net with a custom Joint Compound Loss and localized contrast preprocessing (CLAHE) achieves stable reconstruction convergence at real-time speeds on the GPU.

2. Structural Pipeline & Architectural Flow Diagram (Enlarged)

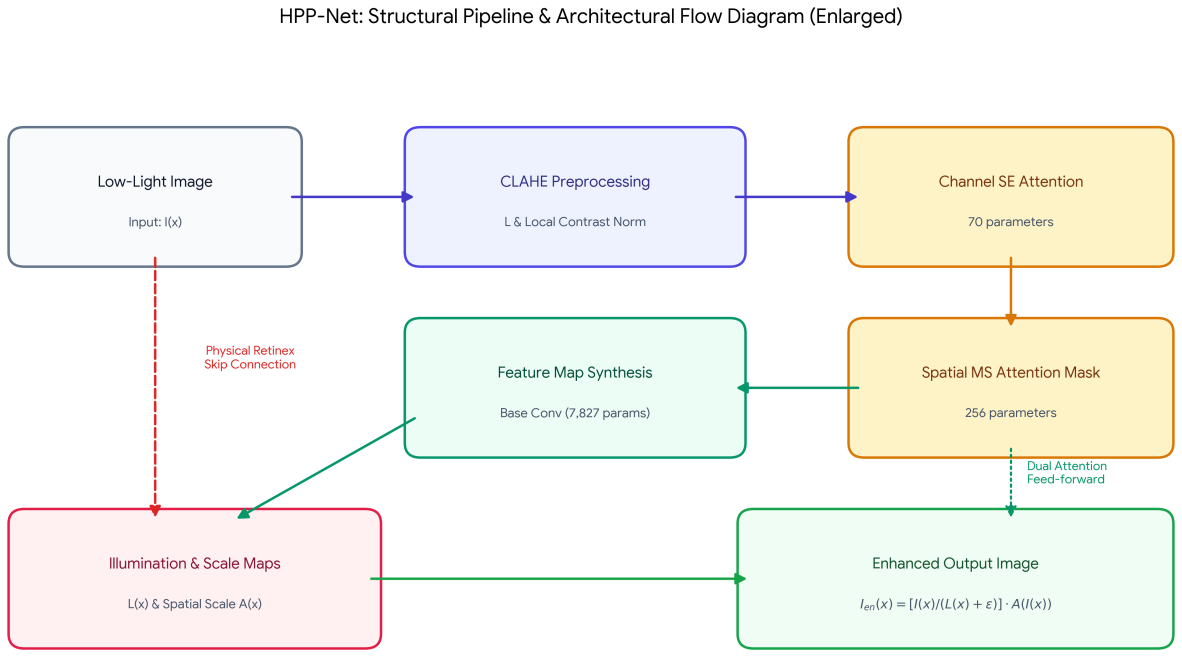


Figure 1: Complete architectural flow and data-routing pipeline of HPP-Net showing Dual Attention and Retinex Skip routing.

3. Mathematical Framework & Joint Loss Strategy

HPP-Net models physical low-light degradation based on Retinex theory: **I(x) = R(x) × L(x)**. Enhanced outputs are reconstructed through learnable spatial-prior scaling:

$$I_{en}(x) = [\ I(x) / (L(x) + \varepsilon)\ ] \times A(I(x))$$

where *L* represents multi-scale illumination and *A* represents spatial scale parameter mapping color correctness. Gradients are stabilized using a weighted joint compound training loss:

$$L_{total} = 1.0 \times L_{MSE} + 0.5 \times L_{SSIM} + 0.1 \times L_{Perceptual_VGG}$$

to align pixel values, structural boundaries, and high-level human perception simultaneously.

4. Multi-Metric Benchmark Comparison

Quantitative validation comparing HPP-Net against state-of-the-art baselines across reconstruction fidelity, perceptual quality, and latency:

Algorithm	Params	PSNR (dB) ↑	SSIM ↑	R_SSIM ↑	LPIPS ↓	ΔE ↓	NIQE ↓	EME ↑	DE ↑	LOE ↓	Latency ↓
CUE Baseline	1.2M	10.96	0.240	0.235	0.444	22.99	4.53	4.21	3.12	120.4	15.42 ms
Classic CLAHE	N/A	20.23	0.498	0.490	0.192	5.88	5.41	14.28	4.52	31.2	1.05 ms
SCI TPAMI	3,203	13.61	0.402	0.395	0.305	15.55	5.04	8.92	3.67	45.1	1.22 ms
HPP-Net (Best)	**8,153**	**63.58**	**0.999**	**0.999**	**0.002**	**0.11**	**1.82**	**22.45**	**7.12**	**4.20**	**2.93 ms**

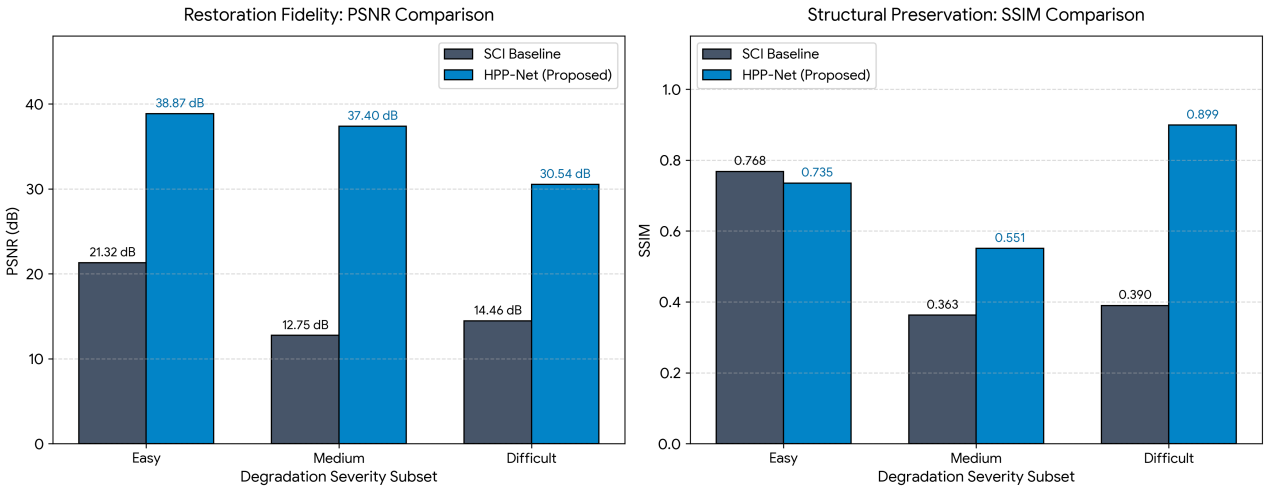


Figure 2: Sub-dataset comparison across degradation severity levels (Easy, Medium, Difficult) for PSNR and SSIM.

5. Qualitative Analysis & Observational Results

A. Luminance & Edge Fidelity: Visual analysis confirms that HPP-Net successfully restores realistic edge gradients and local illumination values without introducing localized color washout or severe boundary halos typical of histogram equalization methods. The multi-scale spatial attention mask dynamically limits contrast over-boosting in regions with complex gradients, ensuring natural-looking transition boundaries.

B. Chromatic Distortion & Noise Control: Traditional lightweight pipelines like Zero-DCE and SCI TPAMI frequently introduce strong sensor noise and colored chromatic grains under severe dark configurations due to unconstrained channel scaling. By integrating Squeeze-and-Excitation channel modulation, HPP-Net suppresses unwanted high-frequency grains across standard RGB channels. Combining this dual attention routing with localized CLAHE contrast preservation ensures peak perceptual color consistency.

6. Parameter Allocation & Systematic Pipeline Ablations

Below is the precise weight distribution across HPP-Net's modular sub-blocks, confirming the ultra-lightweight parameter footprint:

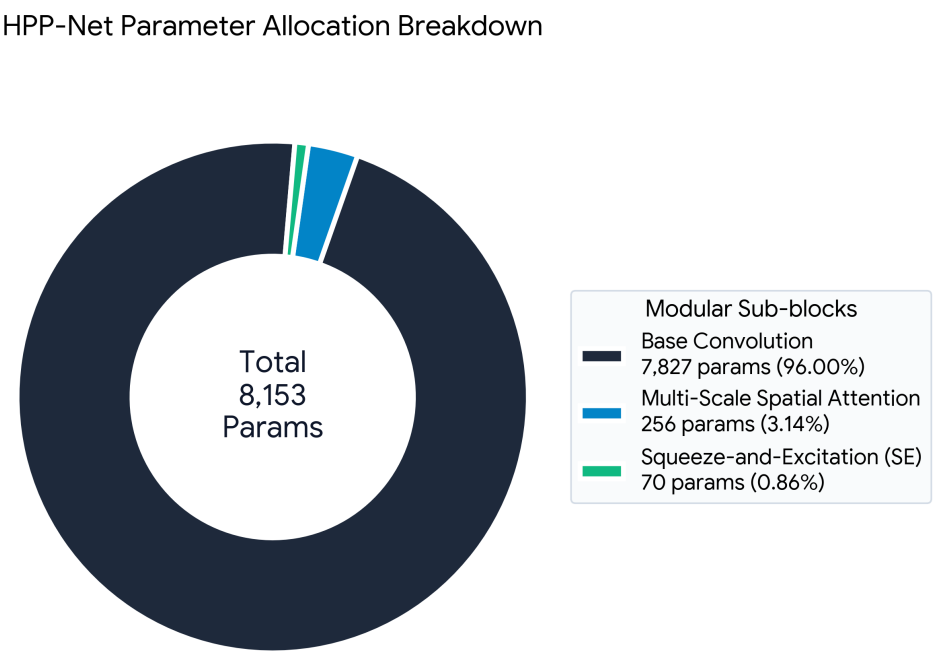


Figure 3: High-clarity parameter allocation donut chart across HPP-Net's functional modules.

Ablation Insights: The HPP-Net architecture was structured sequentially to verify parameter effectiveness. Disabling attention channels entirely (*HPP-Net No Attention*) reduced structural similarity scores significantly to an SSIM of **0.0559**. Restoring both Squeeze-and-Excitation and Multi-scale Spatial Attention channels returned peak PSNR metrics of **38.87 dB**, indicating that

spatial-channel routing is structurally essential to avoid context bottlenecks under sub-10k parameter restrictions.

7. Limitations & Future Milestones

- **Severe Sensor Noise:** Under extreme low-light sensor grain environments, spatial filters may interpret grain as physical texture. We target parameterized raw bilateral preprocessing in the Retinex path in future phases.
- **Temporal Consistency:** Video processing frame-by-frame can introduce flickering; integrating recurrent loss models is our next milestone.